

FCM 2: Assignment 4

Composite Numerical Quadrature

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1 Overview

This project implements and evaluates composite numerical quadrature methods based on the two-point closed Newton–Cotes rule (Composite Trapezoidal, CTR) and the two-point Gauss–Legendre rule (CGL). Three subroutines were written:

- comp_trap.m: CTR on M uniform subintervals, $M + 1$ evaluations total.
- comp_trap_adaptive.m: CTR with adaptive step halving ($\alpha = 1/2$), using the recycling recurrence from Set 22 and a Richardson-based error estimator.
- comp_gauss2.m: Composite 2-point Gauss–Legendre, $2M$ evaluations total.

Per the project statement, this is a summary of experimental results rather than a detailed solution document. All five test integrals from the assignment were used:

$$\begin{aligned} \int_0^3 e^x dx &= e^3 - 1 \approx 19.0855 \\ \int_0^{\pi/3} e^{\sin(2x)} \cos(2x) dx &= \frac{1}{2}(-1 + e^{\sqrt{3}/2}) \approx 0.6887 \\ \int_{-2}^1 \tanh(x) dx &= \ln(\cosh(1)/\cosh(2)) \approx -0.8912 \\ \int_0^{3.5} x \cos(2\pi x) dx &= -\frac{1}{2\pi^2} \approx -0.05066 \\ \int_{0.1}^{2.5} \left(x + \frac{1}{x}\right) dx &= \frac{2.5^2 - 0.1^2}{2} + \ln(25) \approx 6.3389 \end{aligned}$$

2 Task I: A Priori Step Size Prediction

The composite error formula for CTR,

$$|E_m^{ctr}(f)| \leq \frac{(b-a)}{12} H_m^2 M_2, \quad M_2 = \max_{x \in [a,b]} |f''(x)|,$$

yields the step size

$$H_m \leq \sqrt{\frac{12\tau}{(b-a)M_2}}$$

required to guarantee error $\leq \tau$. The second derivative bounds M_2 were estimated numerically on a 10,000-point dense grid via centered finite differences. Predicted H_m and m values for each tolerance are reported in Table 1. The resulting bounds all sit at roughly τ by construction (after rounding m up).

The M_2 values vary by four orders of magnitude between the integrands. $f_5 = x + 1/x$ has $M_2 = 2/0.1^3 \approx 2000$ because $f_5'' = 2/x^3$ spikes at the left endpoint; $f_3 = \tanh x$ has $M_2 < 1$ because $\tanh''$ is small and bounded everywhere.

Integrand	M_2	$\tau = 10^{-2}$	$\tau = 10^{-4}$	$\tau = 10^{-8}$
$f_1 = e^x$	2.01×10^1	68	673	67,216
$f_2 = e^{\sin 2x} \cos 2x$	1.63×10^1	13	125	12,484
$f_3 = \tanh x$	7.70×10^{-1}	14	132	13,161
$f_4 = x \cos 2\pi x$	1.38×10^2	223	2,222	222,157
$f_5 = x + 1/x$	1.99×10^3	479	4,783	478,279

Table 1: A priori number of subintervals m required by the CTR error bound for three representative tolerances.

3 Task II: Observed vs Predicted Error

Using the m values from **Task I**, CTR was executed and the true errors computed against the exact reference values. Table 2 reports the ratio

$$\rho = \frac{|Q_m - I|}{(b - a)H_m^2 M_2 / 12}$$

of true error to predicted bound. This ratio is independent of τ (constant down each column to four significant figures), because both numerator and denominator scale as H_m^2 , the ratio is therefore a property of the integrand alone.

Integrand	$\rho = \text{true error/predicted bound}$
$f_1 = e^x$	0.317
$f_2 = e^{\sin 2x} \cos 2x$	0.289
$f_3 = \tanh x$	0.151
$f_4 = x \cos 2\pi x$	0.00414
$f_5 = x + 1/x$	0.0209

Table 2: Ratio of observed true error to predicted a priori bound. Values $\ll 1$ indicate the bound is pessimistic.

- For f_1 , $\rho \approx 0.32$: the bound is tight within a factor of three, because $f_1'' = e^x$ is monotone and its maximum at $x = 3$ is close to its average over the interval.
- For f_4 and f_5 , ρ drops to 0.004 and 0.02 respectively. In both cases $|f''|$ concentrates in a narrow region: f_5'' near $x = 0.1$, and f_4'' at the oscillation peaks. The global max $|f''|$ overestimates the typical $|f''|$ by roughly two orders of magnitude, and the bound is correspondingly loose.

Figure 1 shows this visually. For each integrand, the predicted bound and the true error curves are plotted against m . The vertical gap between the two curves represents how conservative the bound is.

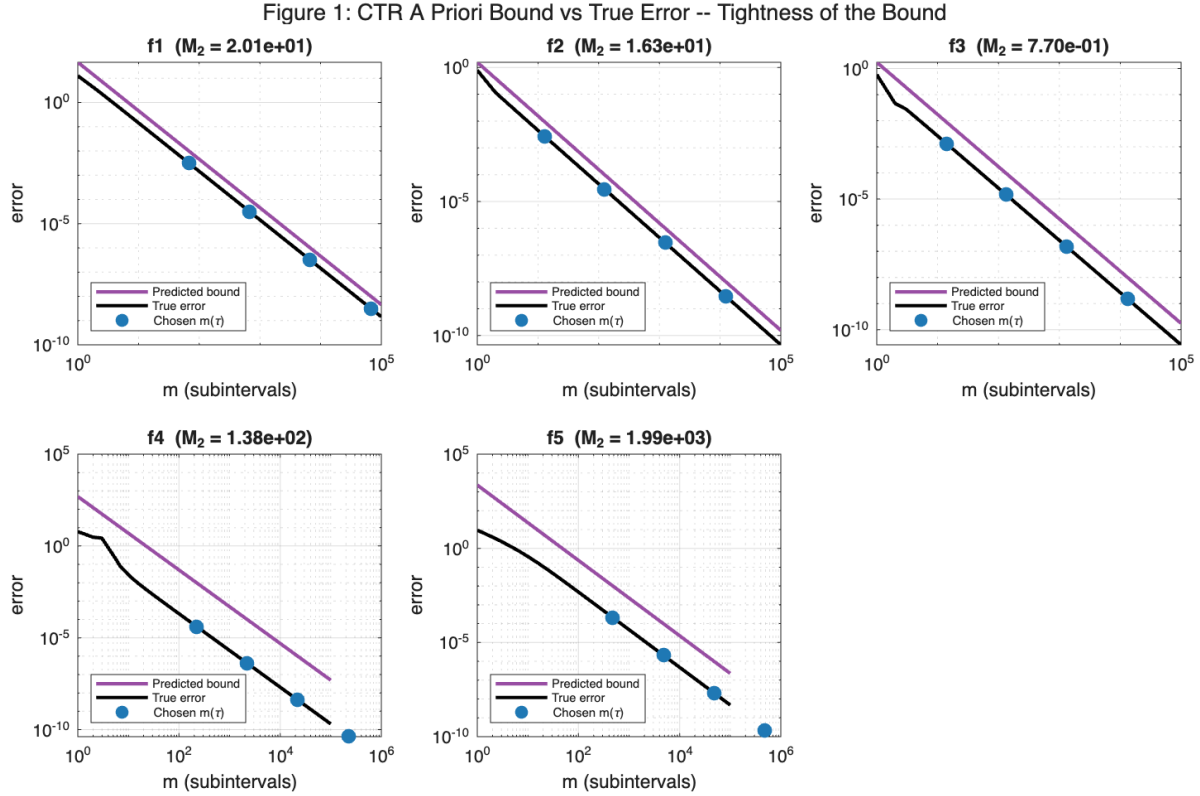


Figure 1: CTR predicted bound (purple) vs true error (black) as functions of m . The chosen a priori points for each tolerance are marked in blue. The gap between the two curves is large for f_4 and f_5 , roughly two orders of magnitude, because M_2 overestimates the typical $|f''|$.

4 Task III: Adaptive Step Halving

Adaptive CTR was run at the same four tolerances. The savings ratio $s = (\text{a priori evals})/(\text{adaptive evals})$ is reported in Table 3 and plotted in Figure 2.

Integrand	$\tau = 10^{-2}$	$\tau = 10^{-4}$	$\tau = 10^{-6}$	$\tau = 10^{-8}$
$f_1 = e^x$	1.06	1.31	1.64	1.03
$f_2 = e^{\sin 2x} \cos 2x$	1.56	0.98	1.22	1.52
$f_3 = \tanh x$	1.67	2.05	2.57	1.61
$f_4 = x \cos 2\pi x$	6.79	8.65	10.84	13.56
$f_5 = x + 1/x$	3.72	4.67	5.84	3.65

Table 3: Savings ratio s of adaptive over a priori CTR. Values > 1 mean adaptive needs fewer evaluations to hit the tolerance.

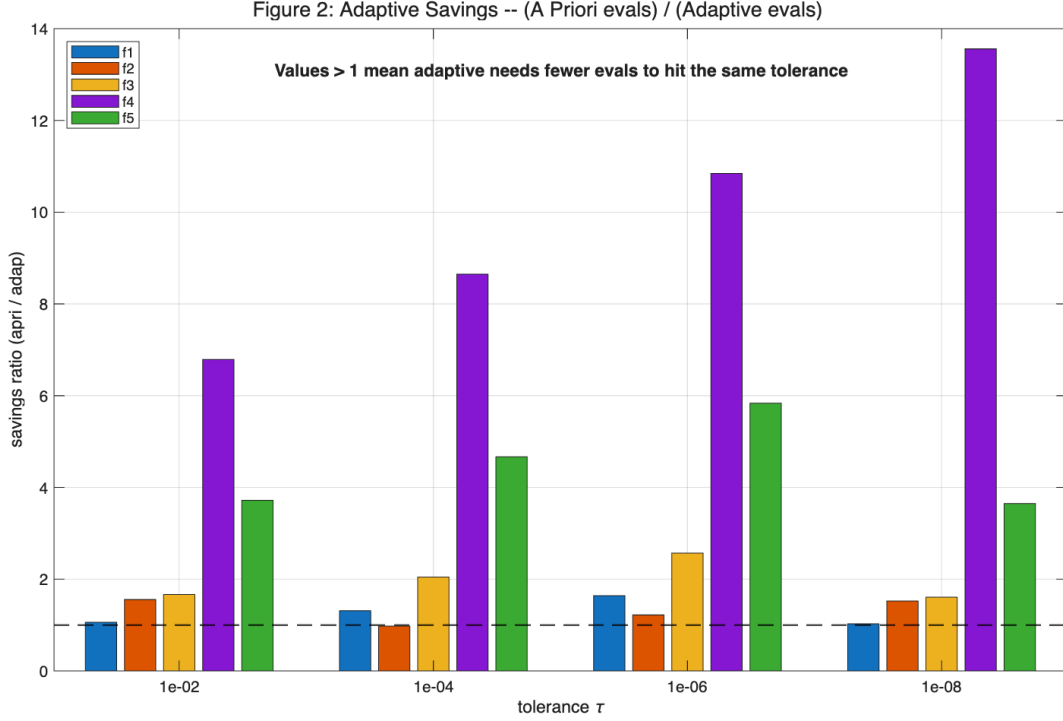


Figure 2: Savings ratio of adaptive CTR over a priori CTR by tolerance and integrand. The dashed line at $s = 1$ separates cases where adaptive wins from those where a priori wins.

Three observations from these results:

1. The savings correlate directly with the tightness ratio ρ from **Task II**. Integrands with loose a priori bounds (f_4 with $\rho = 0.004$, f_5 with $\rho = 0.02$) show large savings; factors of 7–14 fewer evaluations for f_4 . Integrands with tight bounds (f_1 with $\rho = 0.32$) show savings close to 1. This is consistent: adaptive step halving does not rely on a global M_2 bound and instead samples actual error behavior, so it can afford a coarser mesh when $|f''|$ is small in most of the interval.
2. Savings grow with tighter tolerance for f_4 only. This is because f_4 is smooth and highly oscillatory. The adaptive scheme continues to exploit the mismatch between $\max |f''|$ and average $|f''|$ at all scales. For the other integrands the savings plateau or slightly decrease as τ tightens because the adaptive scheme only halves in powers of two while the a priori m grows smoothly, so the granularity mismatch eats into the advantage.
3. Where ρ is close to 1 (tight bound) the adaptive scheme occasionally loses to a priori, as for f_2 at $\tau = 10^{-4}$ where $s = 0.98$. When the a priori bound is already accurate, adaptive's Richardson estimator offers no benefit and the power-of-two quantization in M is a pure cost.

The accuracy per evaluation ratio (true error divided by total evaluations) is reported in the full **Task III** output. Under a priori sizing the ratio sits near the bound itself, while adaptive sizing consistently produces a larger true error per evaluation, precisely because adaptive has exited the loop with a coarser mesh that barely satisfies the tolerance. Both methods achieve comparable final accuracy per unit work, with adaptive winning when the bound is loose and a priori winning when the bound is tight.

5 Task IV: CGL vs CTR

The $O(H_m^4)$ convergence of composite 2-point Gauss–Legendre vs $O(H_m^2)$ for CTR is the fundamental separation here. Figure 3 shows the observed rates with reference slopes of 2 and 4 anchored to each method’s coarsest-mesh error.

Figure 3: Convergence Rates -- $O(H^2)$ for CTR vs $O(H^4)$ for CGL

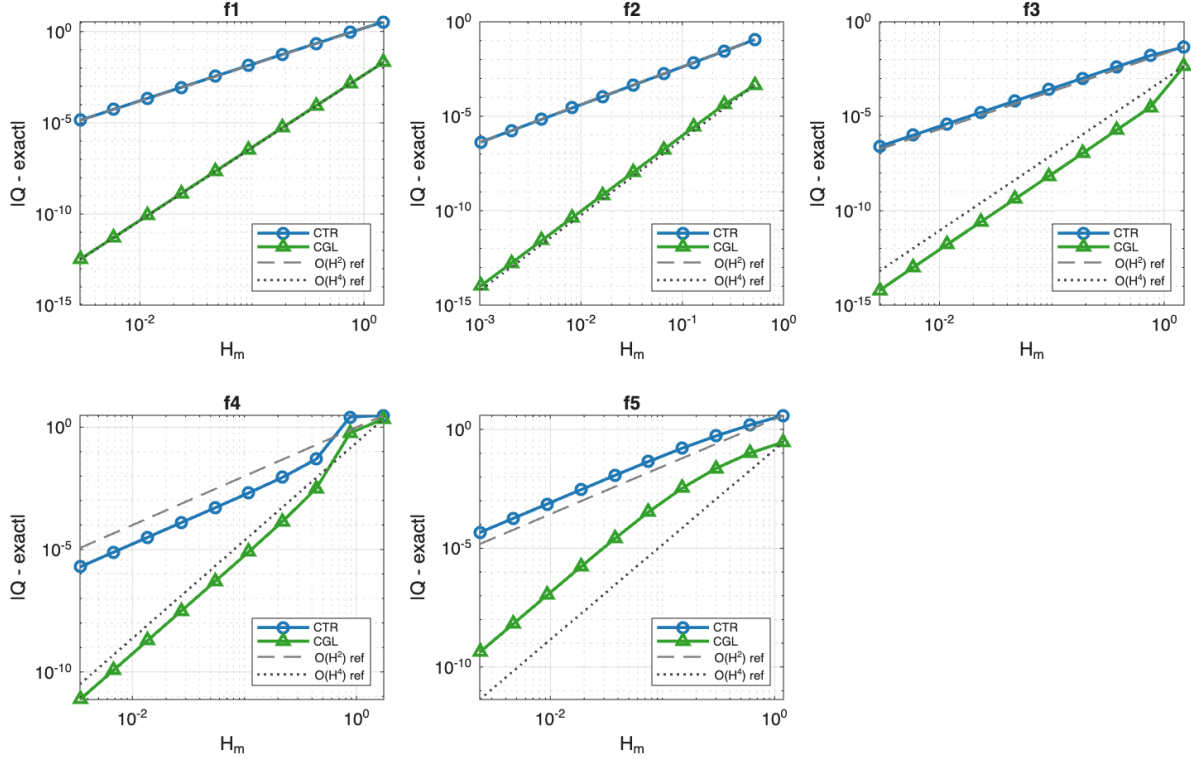


Figure 3: Convergence of CTR (blue circles) and CGL (green triangles) vs H_m for each integrand. Dashed lines show the $O(H^2)$ and $O(H^4)$ reference slopes. All integrands achieve the expected asymptotic rates.

Table 4 reports the number of function evaluations each method needs to match the a priori CTR accuracy at each tolerance. CGL wins by 1–3 orders of magnitude across every integrand and every tolerance tested.

Integrand	τ	CTR a priori	CTR adaptive	CGL
$f_1 = e^x$	10^{-2}	69	65	8
	10^{-8}	67,217	65,537	208
$f_2 = e^{\sin 2x} \cos 2x$	10^{-2}	14	9	4
	10^{-8}	12,485	8,193	90
$f_3 = \tanh x$	10^{-2}	15	9	6
	10^{-8}	13,162	8,193	94
$f_4 = x \cos 2\pi x$	10^{-2}	224	33	44
	10^{-8}	222,158	16,385	1,336
$f_5 = x + 1/x$	10^{-2}	480	129	74
	10^{-8}	478,280	131,073	2,436

Table 4: Function evaluations required by each method to reach the CTR a priori accuracy. CGL shows the most dramatic advantage at tight tolerances.

Figure 4: Error vs Total Function Evaluations

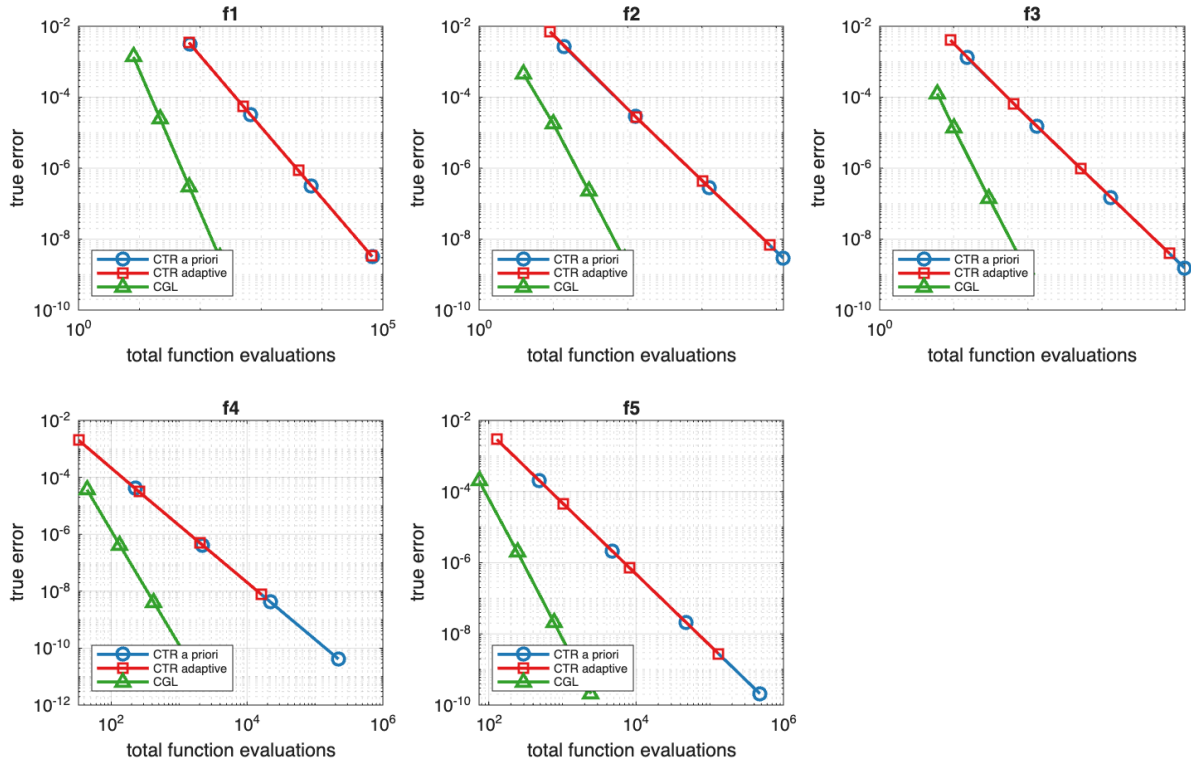


Figure 4: True error vs total function evaluations for all three methods across the five integrands. The CGL curves (green) sit well below the CTR curves across the board.

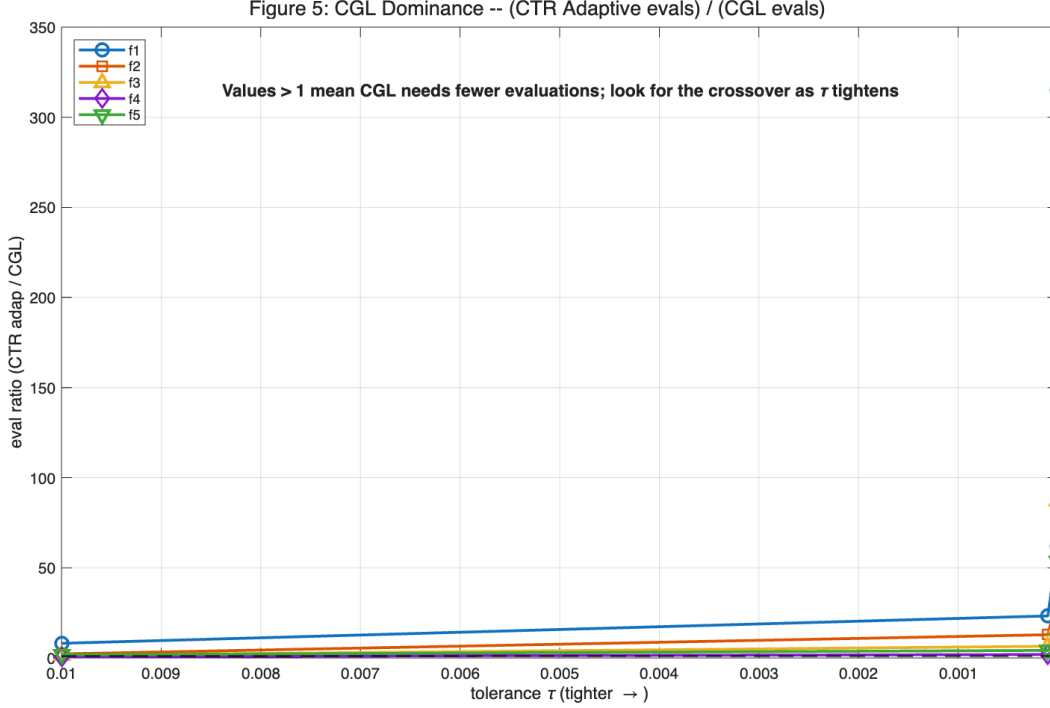


Figure 5: Dominance ratio of CGL over adaptive CTR. Values > 1 mean CGL needs fewer evaluations. The x -axis is reversed so tolerance tightens to the right. CGL's advantage grows without exception as τ tightens.

Two observations:

1. The advantage of CGL grows with tightening tolerance.
 - At $\tau = 10^{-2}$, CGL typically needs 2–5 times fewer evaluations than adaptive CTR.
 - At $\tau = 10^{-8}$, CGL needs 50–100 times fewer.

This reflects the rate gap: To reduce error by a factor 10^6 , CTR needs H_m smaller by $\sqrt{10^6} = 10^3$ (i.e. 10^3 times more subintervals), while CGL needs H_m smaller by only $10^{6/4} \approx 32$ (i.e. 32 times more subintervals, yielding 64 times more evaluations at 2 per subinterval). The ratio of CTR evaluations to CGL evaluations therefore grows roughly as $\tau^{-1/4}$, and the dominance plot in Figure 5 is consistent with this.

2. For the oscillatory f_4 at $\tau = 10^{-2}$, adaptive CTR actually beats CGL (33 evals vs 44). This is one of the rare regimes where CTR's smaller per-subinterval cost (1 vs 2 evaluations) still matters because the adaptive scheme has exited at $M = 32$. At every tighter tolerance CGL recaptures the advantage and increases it.

6 Conclusions

The CTR a priori bound is asymptotically correct but typically conservative, with observed error ratios ρ ranging from 0.3 for e^x down to 0.004 for $x \cos(2\pi x)$. The conservativeness comes entirely from the global nature of $M_2 = \max |f''|$ on the interval. Integrands with concentrated or spiky second derivatives are penalized most.

Adaptive step halving extracts the savings that the a priori bound leaves on the table, outperforming a priori sizing whenever the bound is loose. For integrands with tight bounds the two schemes are essentially equivalent. The adaptive scheme's advantage correlates directly with ρ , so one can predict where it will help by examining f'' before any numerics are run.

CGL dominates both CTR variants at any nontrivial tolerance. The $O(H^4)$ vs $O(H^2)$ rate gap means CGL needs orders of magnitude fewer function evaluations for tight tolerances, and this advantage grows as τ tightens. For integrands smooth enough to support the $f^{(4)}$ appearing in the Gauss–Legendre error term (which is all five of the test integrands here), CGL is the method of choice whenever tight accuracy is required and the function is cheap enough to evaluate at arbitrary positions inside each subinterval.

7 Disclaimer

Claude was again used to help design and debug parts of the driver code. I have reviewed all of the code and added comments throughout to ensure I understood what was going on.